

$$-\Delta u = f \quad \text{in } \Omega = (0, 1)^2,$$

$$u = 0 \quad \text{on } \partial\Omega,$$

$$f(x, y) = 2\pi^2 \sin(\pi x) \sin(\pi y),$$

Find $u \in H_0^1(\Omega)$:

$$\int_{\Omega} \nabla u \cdot \nabla v \, d\Omega = \int_{\Omega} f v \, d\Omega \quad \forall v \in H_0^1(\Omega).$$